VASTRA: DP-900 Data Architecture

Project-Based Learning (PBL) Report | Core Azure Data Fundamentals

A foundational requirement of the VASTRA platform is the efficient and secure processing of disparate data types. In alignment with Microsoft Azure Data Fundamentals (DP-900), the system implements a polyglot persistence strategy, deliberately separating structured relational data from unstructured binary media to optimize both compute performance and storage costs.

1. Structured Data Management

> **Definition:** Structured data adheres to a rigid, predefined schema (rows and columns) and is managed via Relational Database Management Systems (RDBMS). It is highly organized and easily queried using Structured Query Language (SQL).

Within VASTRA, all core transactional and relationship data is classified as structured. To manage this, the platform utilizes **Azure SQL Database**.

Data Entities	User Profiles, Roles, Product Metadata (prices, IDs, URLs), Wishlist mappings, and System Activity Logs.
Schema & Integrity	The database strictly enforces a **Third Normal Form (3NF)** architecture. Relationships, such as a User adding a Product to their Wishlist, are governed by explicit Foreign Key constraints, preventing orphaned records and maintaining referential integrity.
Operational Benefits	• **ACID Compliance:** Ensures that multi-step database transactions (e.g., creating a user and initializing their preferences) are processed reliably. • **Complex Querying:** Enables deep analytical queries, such as executing `INNER JOIN` operations to fetch all products mapped to a specific user's wishlist instantly.

2. Unstructured Data Management

> **Definition:** Unstructured data lacks a predefined data model or schema. It encompasses raw binary files such as images, videos, or documents, which cannot be efficiently stored or queried in traditional relational tables.

VASTRA is an AI-driven visual discovery platform heavily reliant on user-uploaded outfit imagery. Storing these binary image files (JPEGs/PNGs) within Azure SQL would result in severe database bloat, increased latency, and exorbitant compute costs. Therefore, this data is classified as unstructured and routed to **Azure Blob Storage**.

Data Entities	User-uploaded outfit images, AI-generated style reference images, and platform visual assets.
Storage Mechanism	Images are uploaded directly as binary objects (blobs) into designated Azure storage containers. The data possesses no relational schema; rather, it is identified by unique object URIs.
Operational Benefits	• **Infinite Scalability:** Blob storage horizontally scales to accommodate terabytes of media without requiring database provisioning or indexing. • **Cost Efficiency:** Object storage is significantly cheaper per gigabyte compared to premium relational database storage. • **API Interoperability:** Azure Blob Storage generates direct URLs for each image, allowing the VASTRA backend to pass image links directly to external AI vision models (e.g., Groq) for instant processing without loading the binary file into backend server memory.

3. Architectural Synergy

By cleanly separating these data domains, VASTRA achieves optimal cloud efficiency. The Azure SQL Database remains lightweight and highly responsive for user interactions and authentication routing, while Azure Blob Storage acts as a highly durable, low-cost repository serving the heavy media payloads required by the AI styling engine. This dual-pipeline approach embodies the core architectural principles taught in the DP-900 curriculum.